INCH-POUND

MIL-E-29581/1(AS)
21 JUNE 1991

MILITARY SPECIFICATION SHEET

EARCUP, UNIT ACTIVE NOISE REDUCTION SYSTEM, MH-53E AIRCRAFT

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-E-29581.

INTENDED USE: The attenuation levels provided in Table I were calculated to provide adequate protection for aircrew in an MH-53E aircraft during a typical mission. Attenuation levels may not be sufficient to provide adequate protection for use in other aircraft. The ANR system is designed to be electrically compatible with the aircraft Intercommunication System (ICS) AN/AIC-14A, specifically, the microphone circuitry of Interphone Control Box P/N C-2645D. DC power required for operation is received from the ICS microphone circuit via the aircraft ICS cable, and ANR helmet cable. During an aircraft power interruption, the ANR system will automatically switch to the bypass mode. In the bypass mode ICS audio signals are transferred directly to the earphones, without processing of the signal. Manual bypass of the ANR system is provided via a helmet mounted switch.

APPLICABLE DOCUMENTS:

MIL-C-83513	Connectors, Electrical, Rectangular, Microminiature, Polarized Shell, General Specification for
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-C-22442	Cable assemblies, Radio Frequency (aircraft audio), General Specification For

AMSC N/A FSC 5965

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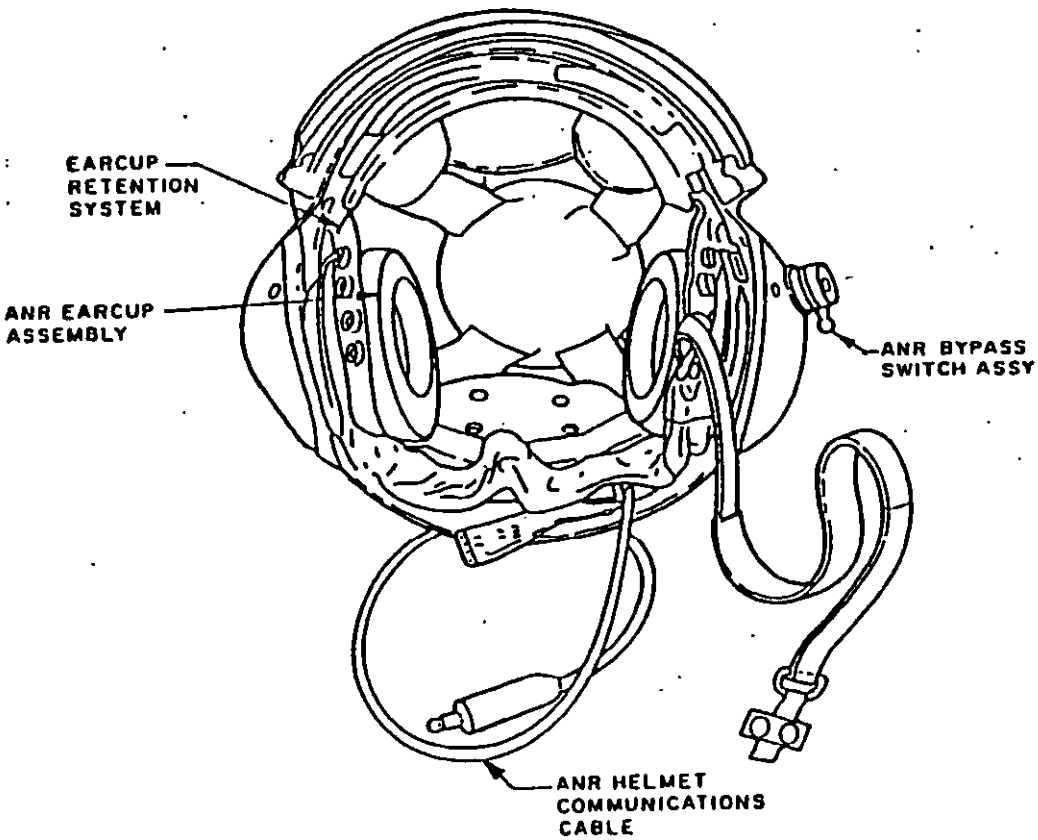


FIGURE 1. Active noise reduction system P/N M29581/1-1 pictorial diagram typical installation assembly.

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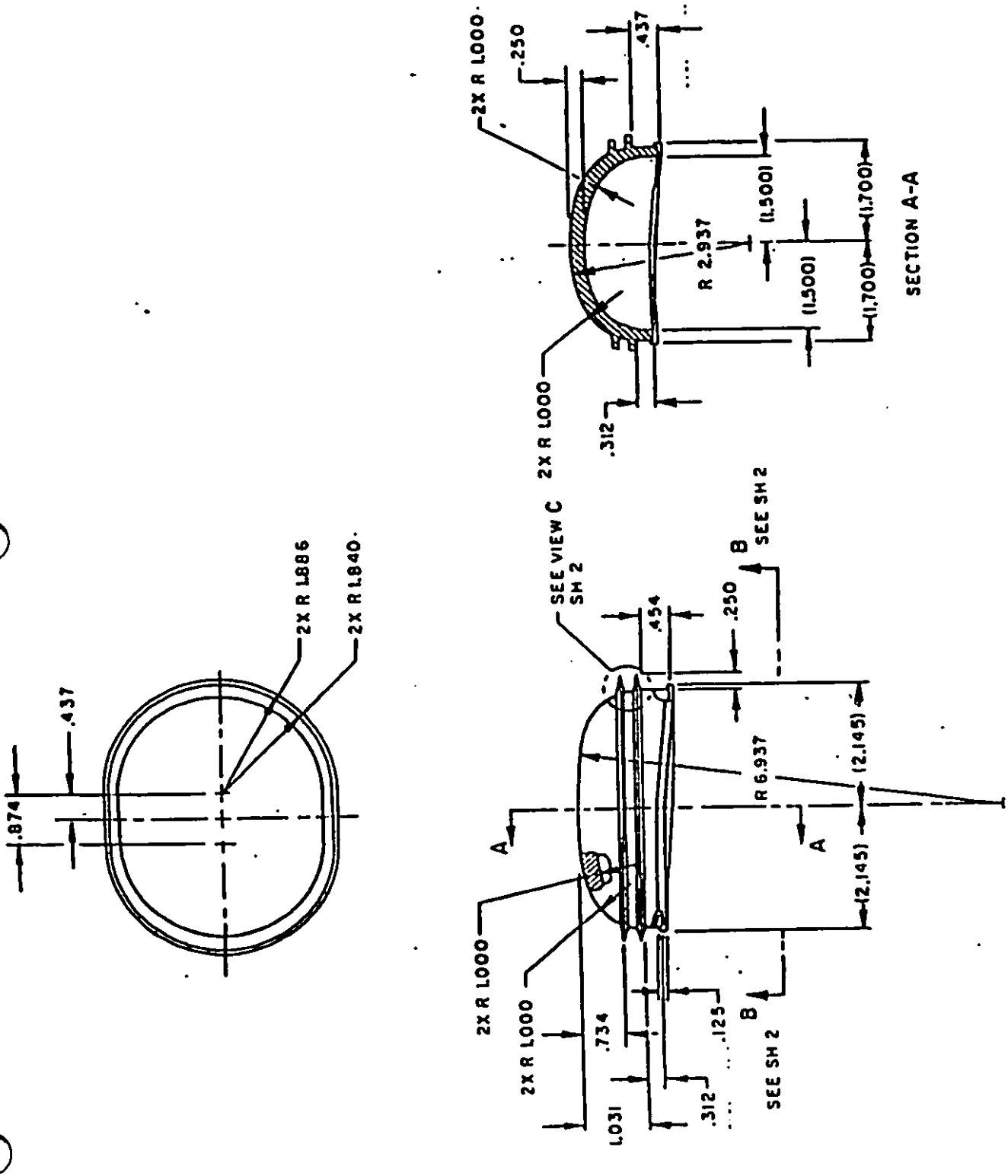


FIGURE 2A. Earcup assembly right P/N M29581/1-2.

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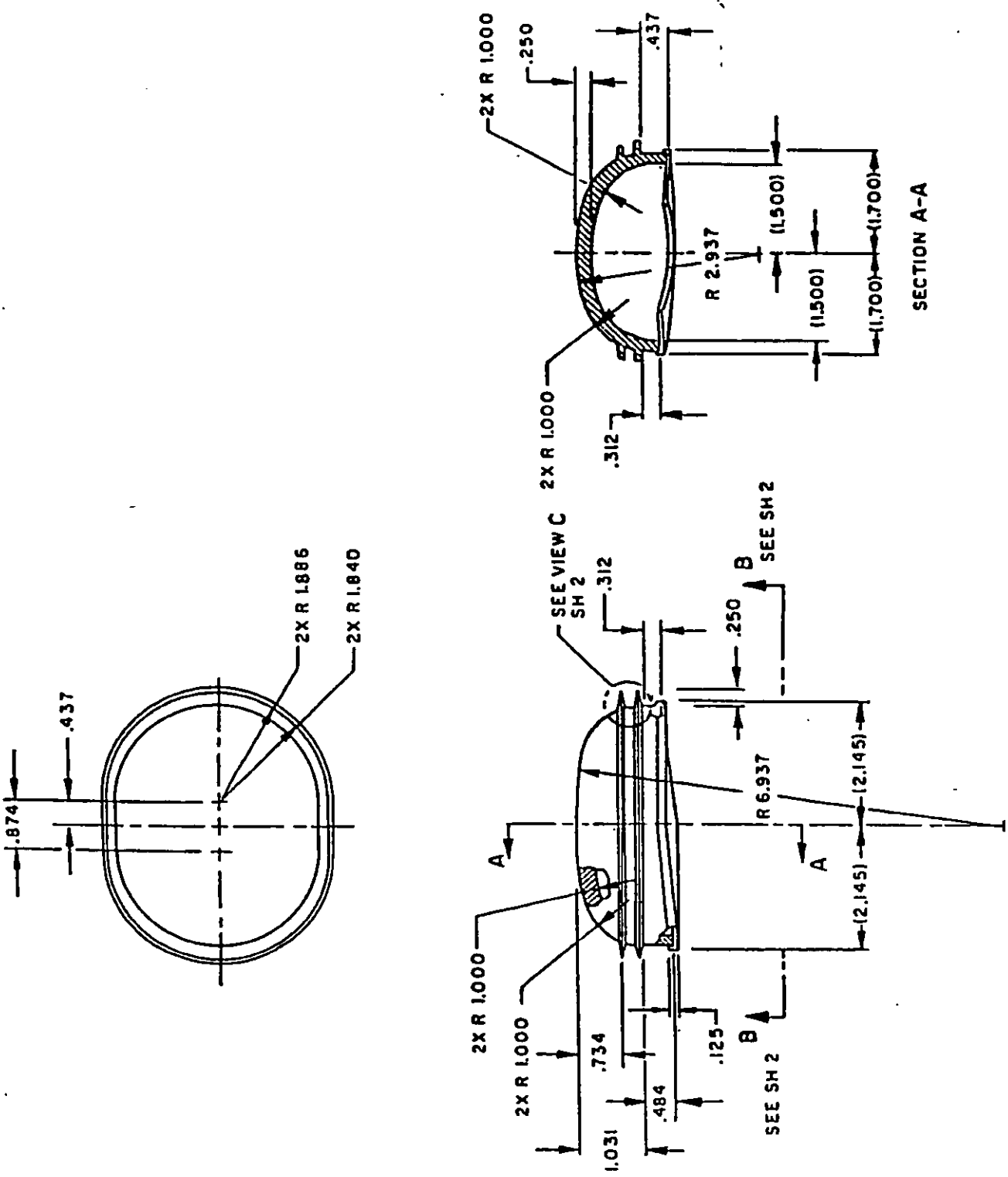


FIGURE 2B. Earcup assembly left P/N M29581/1-3.

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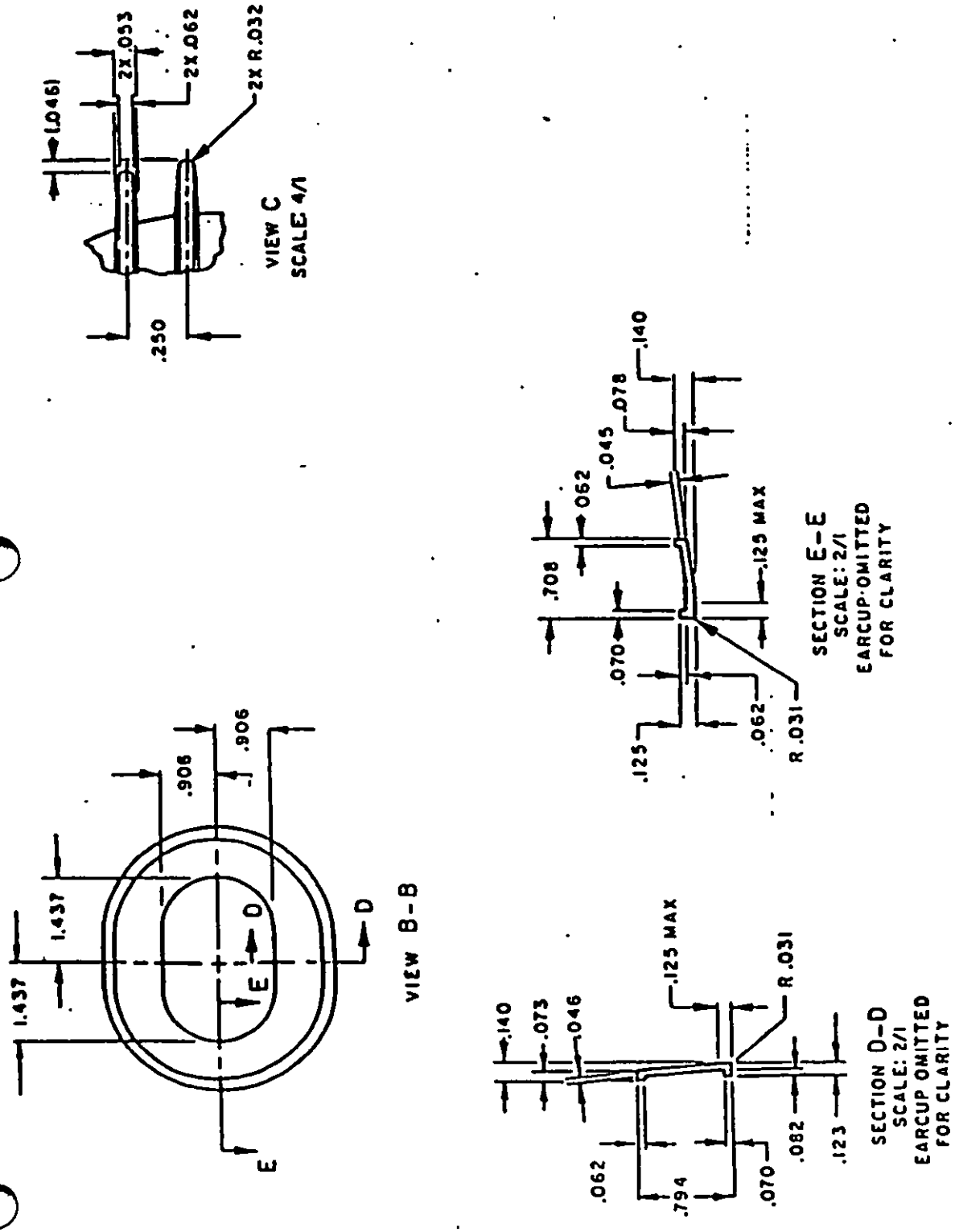
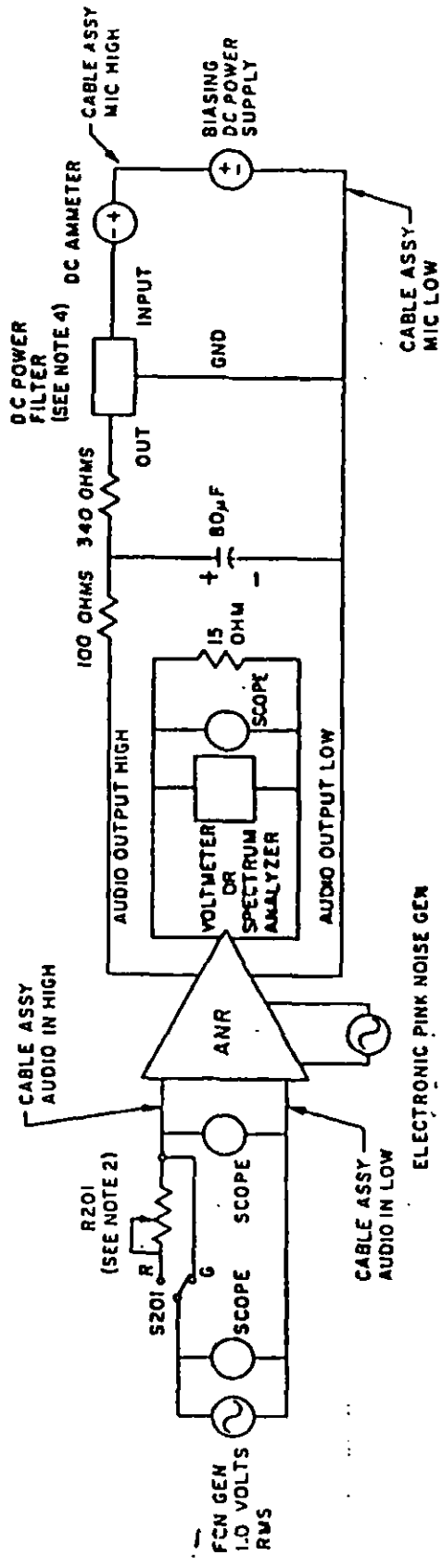


FIGURE 2C. Earcup assembly dimensions.

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NOTES:

1. OSCILLOSCOPE IS REQUIRED TO ASCERTAIN IF ANY WAVESHAPE DISTORTION, ETC. IS PRESENT.
2. R201 DECADE RESISTANCE BOX.
3. SWITCH S201 SHALL BE PLACED IN THE "R" POSITION WHEN PERFORMING THE ANR INPUT IMPEDANCE TESTS AND PLACED IN THE "G" POSITION WHEN PERFORMING ALL OTHER TESTS.
4. THE FILTER P/N IS 18285026-1 AND IS MANAGED BY THE DEFENSE ELECTRONICS SUPPY CENTER DAYTON, OHIO. THE NSN 5915-00-433-3557. THE FILTER IS ADDED TO SIMULATE THE INTERNAL CIRCUITRY OF THE INTERPHONE CONTROL BOX C-2645D. IF A FILTER CANNOT BE OBTAINED, TWO SILICONE DIODES MAY BE USES TO SIMULATE THE VOLTAGE DROP OF THE FILTER. THE POWER SUPPLY SHOULD BE LIMITED TO A MAXIMUM CURRENT OF 40 MILLIAMPERES.
5. TOLERANCES OF RESISTORS ARE 1 PERCENT.

FIGURE 3. Active noise reduction electrical performance and D.C. bias test circuit

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REQUIREMENTS:

1. Passive Attenuation. When tested in accordance with 3.4.1.6.1, the passive attenuation of the helmet-integrated active noise reduction system shall not be less than the values shown below.

Frequency (Hz)	Passive Attenuation (dB)
125	14
250	14
500	24
1000	24
2000	30
3150	40
4000	40
6300	40
8000	40

2. Active and Passive Attenuation (Total Attenuation). When tested in accordance with 4.6.3.6, the combined active and passive attenuation of the helmet integrated active noise reduction system shall not be less than the values indicated below. In addition to the attenuation measurement with a human test subject, the attenuation shall also be measured with an ambient noise level of 125 dB SPL using a manikin.

TABLE I. Active and passive attenuation.

Frequency (Hz)	Combined Active and Passive Attenuation dB SPL
125	18
250	24
500	26
1000	25
2000	30
3150	40
4000	40
6300	40
8000	40

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3. Audio Input Signal. The ANR unit shall accept an audio input signal voltage within the limits of 0.0 to 3.5 volts, RMS, and a frequency range between 300 Hz and 7000 Hz. For testing purposes, the nominal audio input signal shall be 1.0 volt RMS at a frequency of 1000 Hz.

4. Bias Voltages. All tests shall be performed using the circuit shown in Figure 3, with switch S201 placed in the "G" position unless otherwise specified. The DC bias voltage shall be measured at the terminals of the DC supply when configured as shown in Figure 3.

a. The nominal DC bias voltage shall be 28.0 ± 5% volts.
b. The normal DC bias voltage operating range shall be 22.0 to 29.0 volts.
c. A DC bias voltage between 16.0 and 22.0 volts shall be considered a low voltage condition for the ANR system. Low voltage conditions for the microphone is 12.0 to 22.0 volts.

5. Power consumption. The test circuit shown in Figure 3 represents the microphone circuitry of the interphone control box C-2645D, from which the ANR receives DC electrical power. The following conditions apply when connected to the test circuit: When in the bypass mode the ANR system shall not require DC electrical power, except for the microphone. With the DC bias set to 28 volts, the microphone current shall not be greater than 15 milliamperes (mA). When in the active mode, the ANR system shall be capable of meeting all operational requirements specified herein and in the general specification.

6. Impedance With the ANR unit in the bypass mode and referring to Figure 3, set switch S201 so that R201 is out of the circuit. Set the signal generator to a frequency of 1000 Hz, 1.0 volt RMS when measured at the audio input terminals of the helmet cable assembly. Set switch S201 so that R201 is included in the test circuit. Adjust R201 so that the ANR audio input voltage drops to .50 volts RMS as measured by an AC voltmeter. The value of R201 shall be recorded as the ANR input impedance. The audio input impedance of each ANR unit at 1 KHz shall be 15 ohms ± 10% when placed in the active or bypass modes.

7. Earcup dimensions. The earcup assembly dimensions shall be as detailed in Figures 2a, 2b, and 2c, unless otherwise specified in the purchase document for the helmet assembly.

8. Cable assembly. The helmet cable assembly used in the ANR system shall be in accordance with MIL-C-22442/60.

9. Earcup termination. The earcup terminals shall mate with helmet cable connector receptacle P/N M83513/09-A, and mounting jackscrew assembly M83513/5-05 as defined in MIL-C-83513.

10. Environmental conditions. During the performance of the temperature/altitude test, a maximum altitude of 20,000 feet shall be used.

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11. Audio modulation. The DC input power lines of the ANR unit shall be monitored with a spectrum analyzer connected across the 100 ohm resistor shown in Figure 3. The spectrum analyzer shall not indicate presence of a signal greater than 3 millivolts RMS caused by ANR operation in the frequency range specified in 3.4.2.7. A two station AIC-14 ICS system may be used, and a qualitative indication of performance may be substituted for the quantitive measurement defined above. The AIC-14 system shall consist of an interphone control box C-2645 at each station for test subject (A) and (B), and one control box C-4951.

EXPLAINATION OF REQUIREMENTS:

1. Active and Passive Attenuation. The combined active and passive sound attenuation requirements are based on current Navy hearing damage risk criteria for a typical mission in the MH-53E aircraft.

2. Electrical performance. The electrical performance parameters of the ANR unit were chosen to provide compatibility with the aircraft Intercommunication System (ICS) AN/AIC-14A, specifically the Interphone Control Box C-2645D.

3. Audio Signal voltage input range. The audio output signal of the AN/AIC-14A can provide a maximum of 3.5 V RMS into a 7.5 ohm load and 27.0 volts RMS into a 600 ohm load.

4. DC Bias voltage. MIL-STD-704 details the DC characteristics of the aircraft power bus. The ANR units are required to provide the specified performance when subjected to these variations. The ANR unit will receive filtered power from the AN/AIC-14.

Preparing activity
Navy - AS
(Project 5965-N209)

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